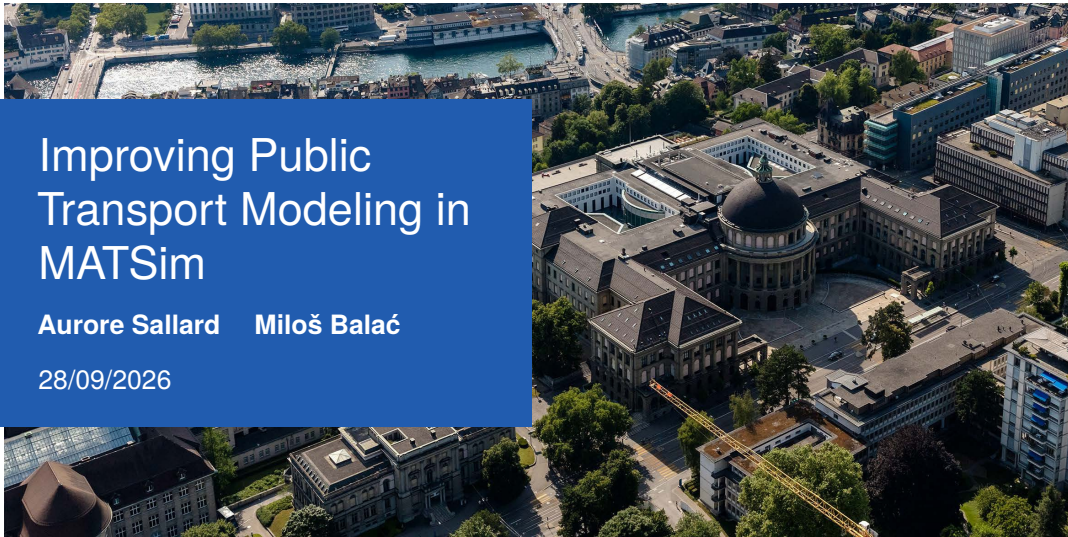


Improving Public Transport Modeling in MATSim

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The CMDP project: a collaborative mobility digital platform

The problem

- MATSim remains largely confined to academia
- Public transport (PT) operators and administrations rely on engineering firms to develop transport models, often based on expensive licensed software building four-step models
- Obstacles to the use of agent-based models: less well-known to practitioners, need for coding in Java to set up and maintain the model

Our goal

- Create a fully automated digital twin of the Swiss mobility based on eqasim and MATSim that can be updated within weeks, and not months or years
- Project partners: Ville de Lausanne, Transports publics lausannois (tl), Transports publics genevois (TPG)
- Supporting institutions: canton de Genève, canton de Vaud, Transitec, Citec, Mobil.home, fondation Modus, Transcality, SDSC

The CMDP project: a collaborative mobility digital platform

Research directions

1. Transport model creation

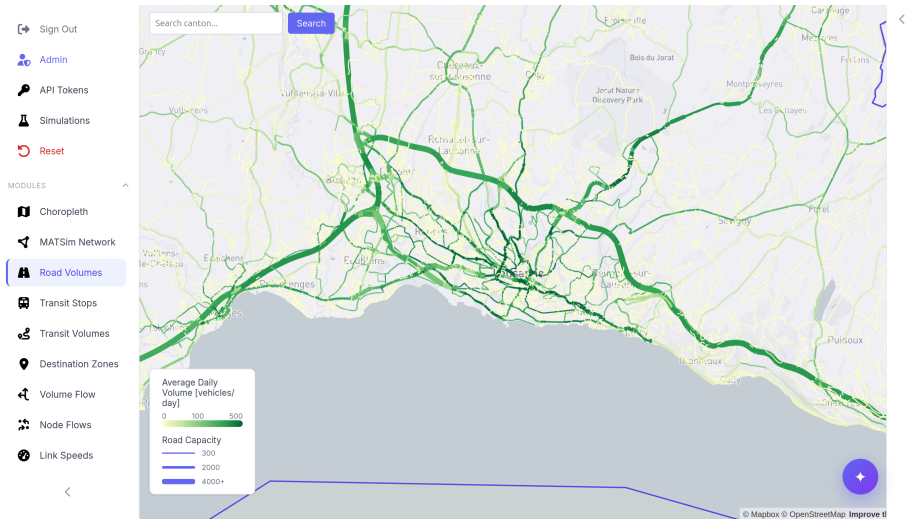
- Improve the eqasim pipeline
 - The eqasim pipeline: flexible framework allowing to generate MATSim input files (transport demand and supply) from raw data [1]
 - Improvements including more detailed synthetic population, addition of cross-border and LCV traffic [2]
- Focus on public transport modelling
- Validate the models on observed data

2. Automated model calibration

- Enable fast, automated, and dynamic calibration
- Presentation by Abdelkader Dib at 09:24, also in room 107 ☺

3. Improving visualization

- Build code-free interactive visualization dashboards for MATSim output data



Visualization example: vehicle counts in the Lausanne region

Contributions

- 1 **A tariff-authority-aware PT fare model** replacing eqasim's default distance-based approximation
 - *Why:* MATSim represents monetary cost only generically (per-distance rate + mode constants), with no notion of an operator's actual pricing rules. Yet fares are a key determinant of PT mode choice and model credibility for policy analysis (e.g. fare reforms)
- 2 **A data-driven calibration of SwissRailRaptor routing parameters** against observed travel behavior
 - *Why:* route choice relies on SwissRailRaptor's generic transfer and waiting penalties, uniform across transfer types. Existing extensions mostly target other aspects (e.g. crowding-dependent disutility).
 - We are specifically interested in intermodality

Outline

1. A tariff-authority-aware fare model

2. Calibration of routing parameters

Swiss PT: A fragmented tariff landscape

Swiss PT is operated by many companies, organized in three broad tiers, each with its own fare logic:

(1) **SBB:** national railway, most regional/long-distance rail

- Distance-based, using standardized tariff distances reconstructed from official reference documents into a weighted graph
- These do not always match real network distances (surcharges on high-demand lines)

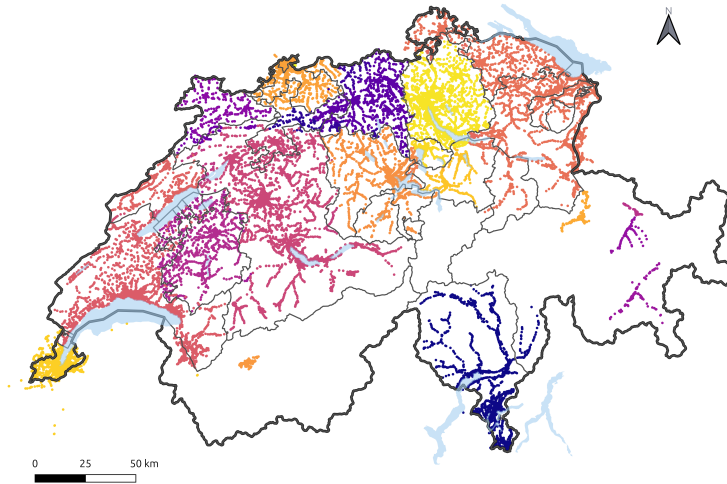
(2) **Regional fare associations:** local operators within an urban region/canton jointly issuing tickets under a common zone-based pricing system

- Priced by number of fare zones crossed
- Structures and pricing rules extracted into a flexible pricing schema

(3) **Independent operators:** local bus companies, mountain railways, each generally setting its own fares

- Too numerous to model individually → single quadratic price-distance model
- Estimated from ~28,700 web-scraped fare observations

Reconstructed coverage: 18 regional fare associations



Runtime fare computation

- Each PT trip's leg sequence is traversed to identify the pricing authority responsible for each segment
- Segment-level information is passed to a fare calculator, which applies the appropriate pricing model per segment to compute the trip's total cost
- This cost can be further adjusted via config-customizable discounts, before being used directly in agents' mode choice utility

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</authority>
```

Outline

1. A tariff-authority-aware fare model

2. Calibration of routing parameters

Calibrating SwissRailRaptor: motivation & method

Motivation: investigate PT route-choice preferences

- Do agents prefer tram over bus? By how much?
- How much extra travel time are agents willing to accept to avoid one transfer?

Method: calibrate via **CMA-ES** (Covariance Matrix Adaptation Evolution Strategy, [3])¹

- Iterative optimization method
- At each iteration:
 1. Evaluate several parameter candidates
 2. New candidates are generated based on the one that performed the best
- Minimizes discrepancy between simulated and observed leg/transfer distributions

Novelty: unlike the reference implementation, we distinguish values of time and transfer penalties by transfer type to capture intermodality

¹<https://github.com/eqasim-org/choice-model>

Data and calibrated parameters

Data: 2015 Swiss Microcensus [4]. The calibration is run on the 16'000 PT trips with walk as access and egress mode. Those trips are characterized by:

- Their number of legs per mode category: train; tram/metro; bus & other
- Their number of transfers per type: train $\leftrightarrow$ train, train $\leftrightarrow$ other, other $\leftrightarrow$ other

Parameters calibrated:

- Mode-specific in-vehicle values of time (train, tram/metro, bus & other)
- Transfer penalties, by transfer type

Initial values: from a SP study conducted by the Swiss Federal Office for Spatial Development (ARE, [5]). PT sub-modes or intermodal transfers were not distinguished, so all parameters start identical

Walk and wait values of time excluded from calibration:

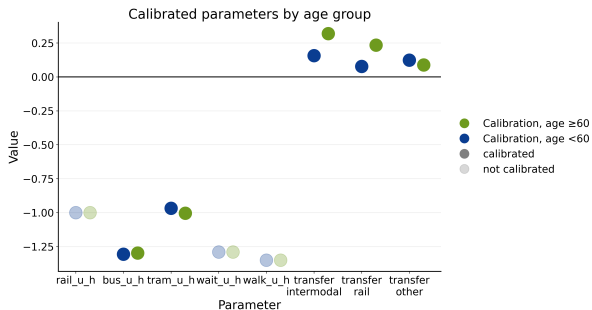
- Walk time imprecisely recorded in the Microcensus
- MATSim's plan optimization leaves agents essentially no waiting time

Calibration results

	Before calibration	After calibration
<i>In-vehicle values of time [h^{-1}]</i>		
train (reference)	-1.00	-1.00
tram + subway	-1.00	-0.95
bus + other	-1.00	-1.29
<i>Other values of time [h^{-1}]</i>		
walk (not calibrated)	-1.35	-1.35
wait (not calibrated)	-1.29	-1.29
<i>Transfer penalties [-]</i>		
train $\leftrightarrow$ train	0.09	0.07
train $\leftrightarrow$ other	0.09	0.16
other $\leftrightarrow$ other	0.09	0.12

- Bus/other: larger in-vehicle disutility $\rightarrow$ lower perceived comfort
- Tram/metro: marginally more attractive $\rightarrow$ higher service quality
- Transfer penalties: train $\leftrightarrow$ train cheapest; intermodal transfers penalized $\sim 2\times$ more

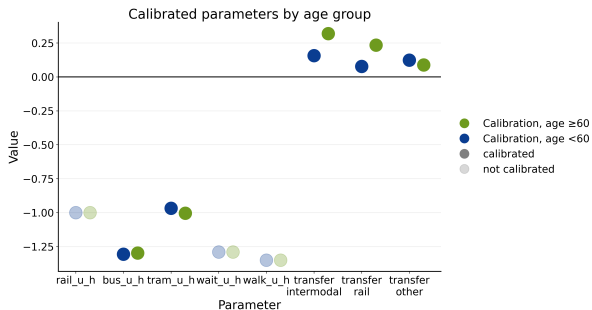
Outlook: towards population-specific route choice parameters



First result: age only

- Calibrated separately for age ≥ 60 and < 60
- In-vehicle values of time are similar in both groups
- Transfer penalties differ markedly, e.g. train $\leftrightarrow$ other: 0.32 vs. 0.16
- $\Rightarrow$ route choice preferences are not homogeneous

Outlook: towards population-specific route choice parameters



Next steps

1. Identify which characteristics drive PT route choice: age, income, health, residence accessibility / land use, ...
2. Cluster the population into meaningful groups and calibrate parameters for each group

Thank you!

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Code: <https://github.com/eqasim-org/eqasim-switzerland/tree/cmdp>
<https://github.com/eqasim-org/eqasim-java/tree/cmdp>

References

- [1] S. Hörl and M. Balac, “Synthetic population and travel demand for Paris and Île-de-France based on open and publicly available data,” Transportation Research Part C: Emerging Technologies, vol. 130, p. 103291, 2021.
- [2] A. Dib, A. Sallard, and M. Balać, “Agent-based transport model of Switzerland based on eqasim and MATSim,” in Proceedings of the 26th Swiss Transport Research Conference (STRC 2026), (Ascona), May 2026.
- [3] N. Hansen, S. D. Müller, and P. Koumoutsakos, “Reducing the time complexity of the derandomized evolution strategy with covariance matrix adaptation (cma-es),” Evolutionary computation, vol. 11, no. 1, pp. 1–18, 2003.
- [4] Federal Office for Spatial Development and Federal Statistical Office, “Verkehrsverhalten der Bevölkerung. Ergebnisse des Mikrozensus Mobilität und Verkehr 2015,” 2017.
- [5] C. Weis, M. Vrtic, B. Schmid, and K. W. Axhausen, “Analyse der SP-Befragung 2015 zur Verkehrsmodus- und Routenwahl,” 9 2017.